

NaijaSense AI Solution Paper

DSN × Bluechip Tech LLM Agent Challenge · DSAS 2026

Team: TAOTECH SOLUTIONS

Abstract

NaijaSense AI is a dual-task, stateful LLM system for **Task A** (review + rating simulation) and **Task B** (personalized recommendation ranking). The core differentiator is silent history retrieval by user_id before generation, followed by persona merge with current user inputs. The stack combines a role-split model strategy (fast router + strong generator), retrieval-augmented review generation, optional critique-regenerate quality control, and a deterministic recommendation scorer with explainability traces.

The system ships with a production-style hub (/unified) that includes single-response UX, live NDJSON reasoning timeline, safety advisories, language control (English/Pidgin/Yoruba mix), health pre-warm status, and thumbs feedback logging. Results show LLM generation is the strongest quality lever for Task A, while Task B ranking remains interpretable but weak on hard same-domain distractors; the next step is LLM reranking on top-K candidates.

1. Problem and Goals

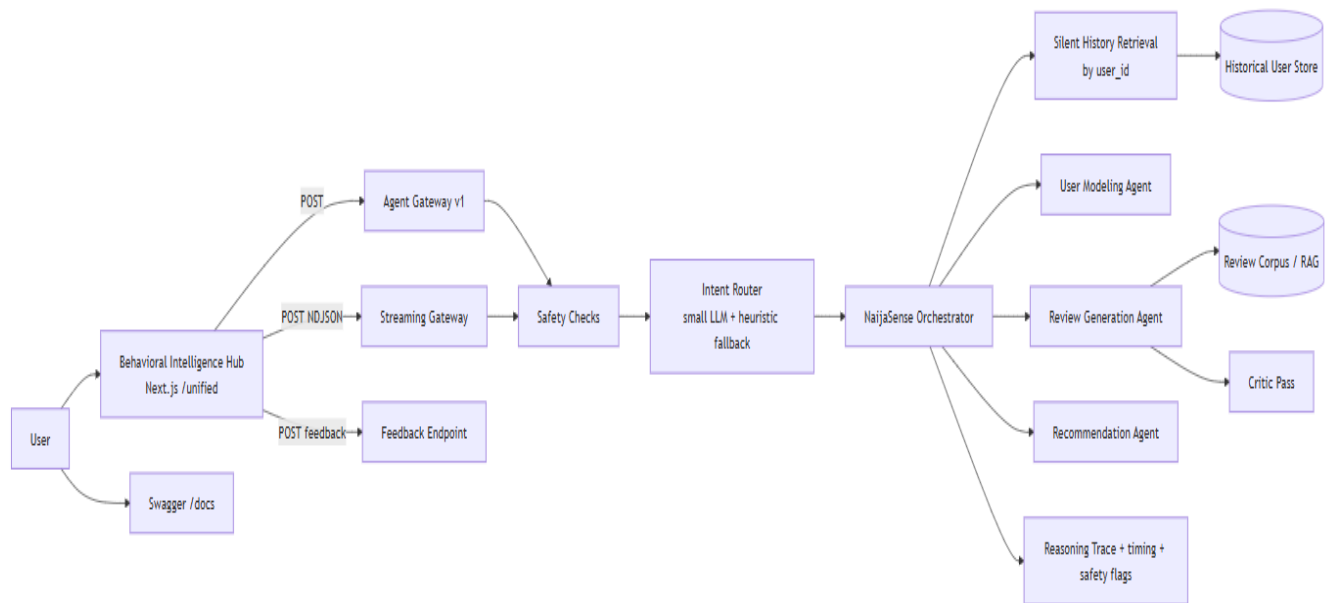
Online reviews encode behavior: tone, rating bias, domain preferences, and context sensitivity. Most baseline systems underperform because they treat users as static profile fields and hide their reasoning.

The system design emphasizes:

- Faithful personalized review simulation (Task A)
- Explainable recommendation ranking (Task B)
- Nigerian context readiness
- Reproducibility and honest reporting

Design constraints: output diversity, reasoning transparency, and low-latency/low-cost deployment.

2. System Overview



2.1 Role-split model strategy

- **Router role** (llama-3.1-8b-instant): intent routing, persona inference, critique scoring.
- **Generator role** (llama-3.3-70b-versatile): review text generation only.

This keeps cost low while preserving quality where it matters most.

2.2 Silent context retrieval (core differentiator)

For every request, the system runs a pre-LLM step:

1. Pull up to five past records for user_id.
2. Build HistoricalPersona (avg_rating, rating_tendency, tone_signal, top domains/interests).
3. Merge with UI persona fields using default-vs-override logic.
4. Log provenance in reasoning metadata.

Unknown users fall back to current input signals.

3. Task A: Review + Rating Simulation

3.1 Generation approach

We use a **facts-in, prose-out** prompt contract:

- Input: item, domain, user persona, optional context.
- Retrieval: top-3 related reviews from corpus as style/concreteness references.
- Guardrail: explicit “do not copy facts from retrieved examples.”

3.2 Diversity controls

To avoid repeated outputs on identical requests, generation uses per-call seed, tuned sampling (temperature, top_p, presence/frequency penalties), and anti-template prompt rules.

3.3 Critique-regenerate pass

A low-cost critic scores review specificity (1-5 rubric). If below threshold, the system regenerates with explicit issue prompts. Most outputs pass in one shot, keeping cost low.

3.4 Language and local context

Output modes:

- english
- pidgin
- yoruba_mix

This supports local contextualization without forcing slang into formal outputs.

4. Task B: Personalized Recommendation

4.1 Deterministic hybrid ranker

Task B ranking is intentionally deterministic and auditable. Score combines:

- interest overlap
- memory overlap
- context overlap
- domain alignment
- rule-based boosts (cold-start, cross-domain, query intent)
- penalties for placeholder-like candidates

The conversational summary is LLM-generated but **does not** alter ranking order.

4.2 Multi-turn behavior

A per-user rolling buffer is threaded into Task B requests. Previous turns influence ranking signals and are surfaced in explainability outputs.

5. Product Surface and Observability

The Behavioral Intelligence Hub is part of the solution quality, not only a demo shell.

Key shipped features:

- Single-response UX (no duplicate compare outputs)
- Live reasoning timeline from /api/agent/v1/stream
- Backend status pill with pre-warm health check
- Routed-task and latency chips
- Safety advisory badges (safety_flags)
- Thumbs feedback to JSONL (/api/agent/feedback)

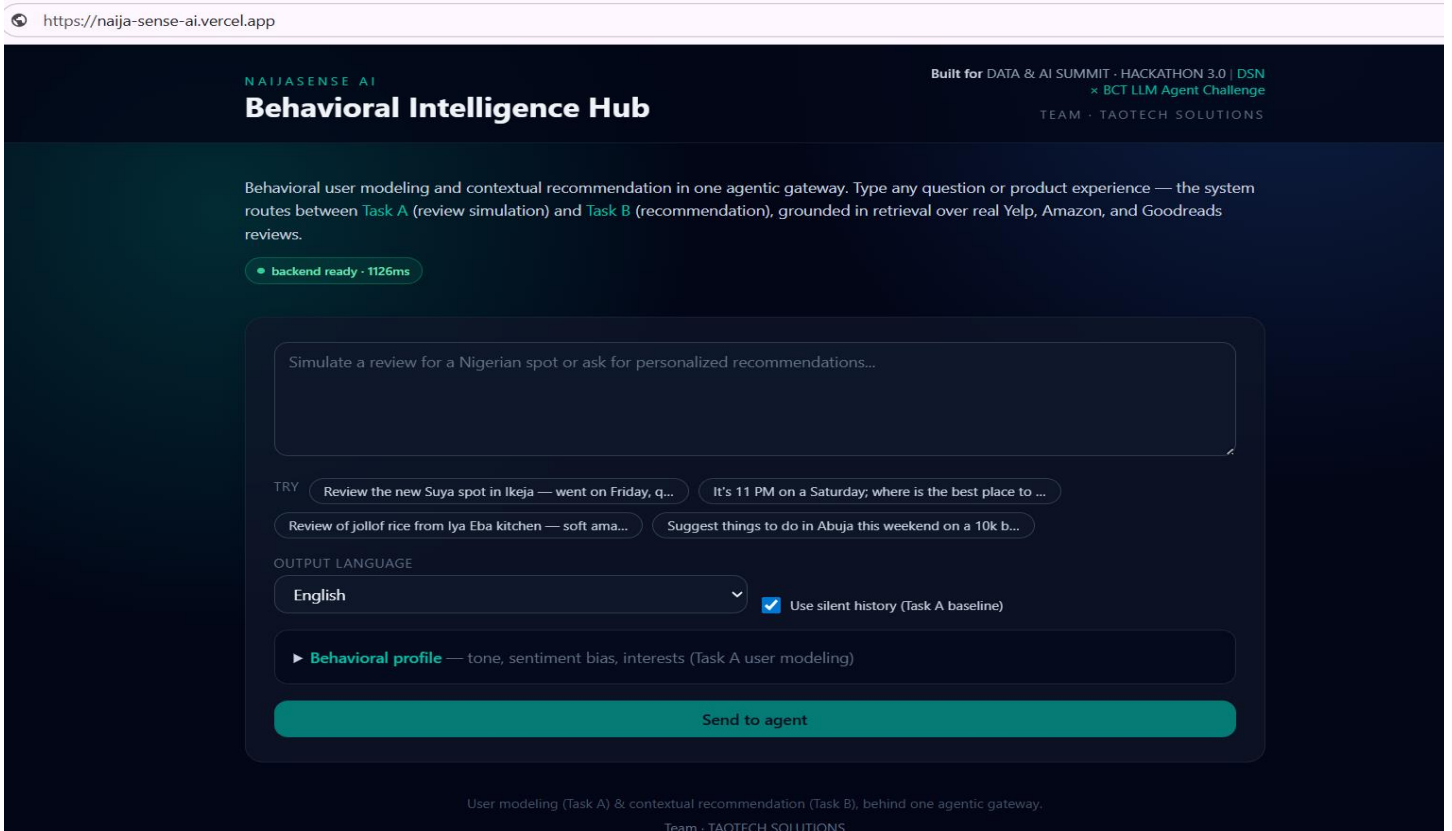


Figure 1. Home screen (homescreen.png): initial view shown after opening <https://naija-sense-ai.vercel.app/>, including quick prompts, language selector, and status indicator.

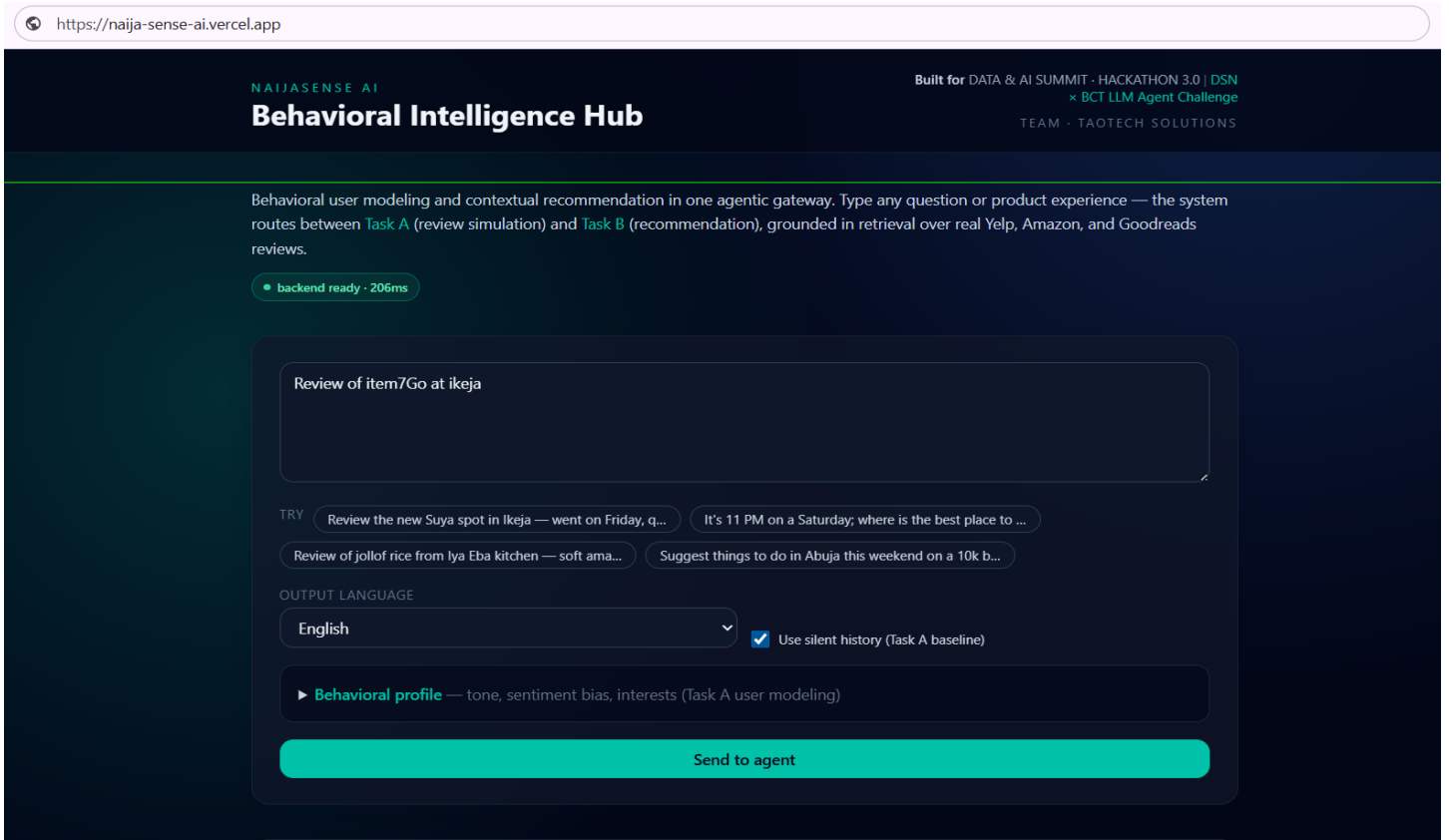


Figure 2. Input workflow (input.png): the interaction step where the user enters query, persona context, and language before submitting to the gateway.

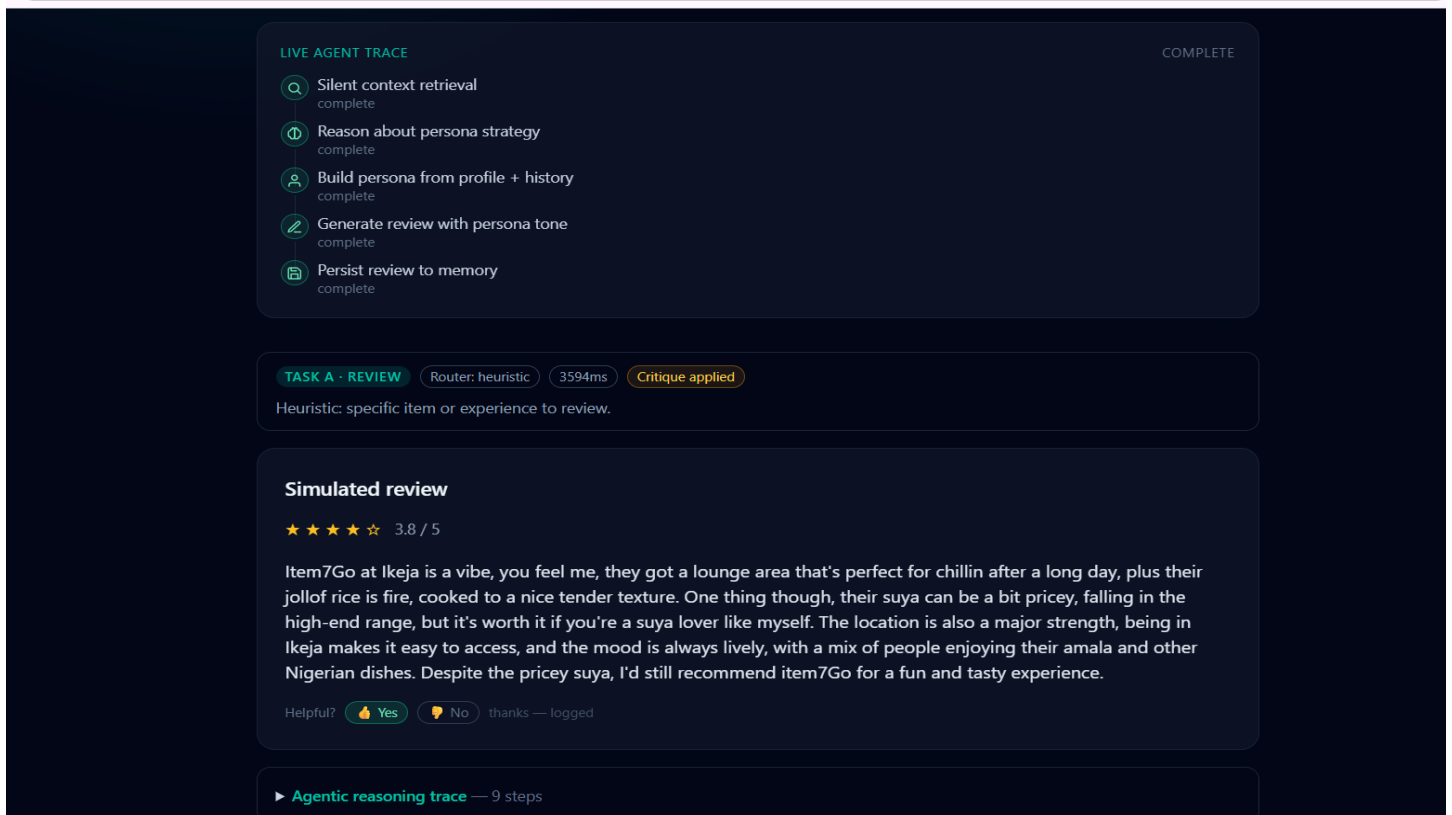


Figure 3. Output view (output.png): result card after inference, showing routed task, generated content, safety advisories, and reasoning trace.

The UI flow is: **Figure 1 (entry)** -> **Figure 2 (input)** -> **Figure 3 (output)**.

6. Experiments and Findings

6.1 Ablation setup

Variants:

Variant	Disabled component
Full rag	none
no_rag	retrieval examples removed
no_critique	critique-regenerate off
no_llm	generation via deterministic fallback only

6.2 Task A results

Variant	ROUGE-1 ↑	ROUGE-L ↑	Token-F1 ↑	RMSE ↓
full	0.161	0.104	0.128	1.251
no_rag	0.187	0.109	0.129	1.003
no_critique	0.165	0.102	0.132	1.240
no_llm	0.126	0.086	0.123	1.242

Interpretation: LLM generation is the strongest quality lever (largest drop in no_llm), RAG can lower lexical overlap while improving concreteness, and critique mainly improves qualitative quality.

6.3 Task B results

Variant	NDCG@10	Hit Rate@10
full	0.062	0.20
no_rag	0.062	0.20
no_critique	0.062	0.20
no_llm	0.062	0.20

Random baseline on this hard set is higher for Hit Rate@10 (~0.50), revealing a ranking gap.

6.4 Behavioral-fidelity A/B

scripts/eval_fidelity.py runs paired tests with and without history (include_history=true/false) and compares rating error, token similarity, and tone match. This isolates the contribution of silent context retrieval.

7. Reproducibility

Environment: Python 3.11+, pip install -r requirements.txt, and .env from .env.example.

Run stack:

```
docker compose up --build
```

Core endpoints: <http://localhost:8000> (API), <http://localhost:8000/docs> (Swagger), <http://localhost:3000/unified> (Hub UI).

Evaluation: `python scripts/run_real_benchmark.py --all_variants` and `python scripts/eval_fidelity.py`.

8. Limitations and Next Steps

1. **Task B reranking:** deterministic scorer needs LLM reranking over top-K to handle subtle same-domain distractors.
2. **Metric depth:** token-F1 fallback may be used where BERTScore installation is constrained; full semantic metrics should be run in a compatible env.
3. **Data coverage:** Amazon slice was smaller than intended in our benchmark run due to external data endpoint availability.
4. **Safety robustness:** regex safety layer is practical but not exhaustive; learned safety critics can improve recall.
5. **Feedback learning loop:** thumbs data is logged; next step is periodic distillation into adaptive few-shot banks.

9. Conclusion

NaijaSense AI delivers a practical, auditable, and locally contextualized dual-task agent for review simulation and recommendation. The strongest contribution is a measurable stateful workflow: silent history retrieval, persona merge, and transparent reasoning traces visible both in API outputs and in the live hub.

Task A quality is competitive and diverse; Task B explainability is strong but ranking on hard distractor sets requires LLM reranking. The system is reproducible, observable, and ready for iterative improvement.